# Capstone Project

# Integrity Risk Assessment Agent

## Retrieval Design for the Integrity Risk Assessment Agent

The Integrity Risk Assessment Agent evaluates whether a company is safe to partner with by grounding its reasoning in real-time evidence, historical semantic memory, and adaptive calibration thresholds. Retrieval is not an optional enhancement but a foundational requirement for the agent's reliability. Integrity risk is dynamic: sanctions lists update frequently, enforcement actions evolve, and reputational signals shift rapidly. A prompt-only model cannot track these changes or learn from past mistakes. Retrieval ensures the agent remains grounded in authoritative data, while the calibration feedback loop ensures it becomes more accurate over time.

### 1. Why Retrieval Is Required

Retrieval is essential for three core reasons.

**First**, the agent must access real-time sanctions lists, regulatory filings, enforcement actions, and news reports. These sources change daily and cannot be encoded in the model's parameters. Retrieval provides the grounding necessary to avoid hallucinations and outdated conclusions.

**Second**, the agent must incorporate historical context to detect patterns such as recurring controversies or risk trajectories. This requires long-term memory that persists across assessments. Semantic retrieval from a vector database allows the agent to recall relevant past events even when phrased differently from the current query.

**Third**, retrieval is required for conflict resolution. When the agent encounters contradictory signals such as a news report claiming a sanction that does not appear in official lists it must retrieve additional evidence to resolve the discrepancy. Without retrieval, the agent would be forced to guess.

Together, these requirements make retrieval a core architectural component rather than an optional feature.

### 2. Integration of Semantic Retrieval Using an External Data Source

The agent uses a **hybrid retrieval architecture** consisting of:

**A. Real-Time Retrieval (External Tools)**

These include sanctions APIs, regulatory databases, news feeds, and entity-resolution services. They provide authoritative, time-sensitive evidence.

**B. Long-Term Semantic Retrieval (Vector Database)**

The agent stores only **meaningful, stable, reusable facts** extracted from past assessments. These are embedded and stored in a vector database to support semantic similarity search. Examples include:

- summaries of past controversies
- historical sanctions events
- prior risk assessments
- conflict-resolution notes
- normalized risk signals (severity, category, timestamp)
- calibration outcomes (false positives, false negatives, borderline cases)

The agent retrieves from this vector store at two points:

1. **At the start of the reasoning loop**, to provide historical context
2. **During conflict resolution**, to check whether similar contradictions occurred before

This retrieval directly influences planning, evidence weighting, and confidence scoring.

### 3. How Retrieval Meaningfully Influences Output

Retrieval is not decorative; it changes the agent's conclusions.

**Example: Sanctions Conflict Scenario**

Suppose the agent receives conflicting signals:

- OFAC sanctions API → *Company X is not sanctioned*
- Reuters news → *Company X sanctioned yesterday*

A prompt-only model might hallucinate a synthesis or choose one source arbitrarily.

With retrieval:

1. The agent retrieves **historical semantic facts**:
    - "Company X was sanctioned in 2022 and removed in 2023."

    - “Company X has recurring corruption allegations.”
    - “Past assessments flagged similar news-vs-API conflicts.”
2. The agent retrieves **additional real-time evidence**:
    - EU sanctions list
    - Regulator press releases
    - Cross-jurisdiction checks
3. The agent weighs:
    - authority (official lists > news)
    - recency (yesterday’s report)
    - historical patterns (recurring issues)
    - calibration thresholds (e.g., past false positives from news-only signals)
4. The final output becomes:
    - “Company X is *likely* sanctioned pending official confirmation.”
    - “Confidence: Medium.”
    - “Reason: Conflicting signals + historical pattern + calibrated weighting.”

Retrieval and calibration together produce a nuanced, defensible conclusion.

## 4. Key Retrieval Design Choices

### A. Data Source Selection

The agent retrieves from:

- official sanctions lists (OFAC, EU, UN)
- regulatory filings (SEC, FCA, etc.)
- enforcement databases
- reputable news sources
- ESG incident databases
- the agent’s own long-term memory (vector DB)

These sources were chosen for authority, update frequency, and relevance.

### B. Document Segmentation and Chunking

For long-term memory, the agent stores **summaries**, not raw documents. Each summary is:

- 200–400 tokens
- focused on a single event or risk signal
- normalized into a consistent schema

This ensures embeddings represent clean, semantically meaningful units.

### C. Number of Retrieved Results

The agent retrieves:

- **Top 5 semantic matches** from the vector DB
- **Top 10 news articles** from real-time retrieval
- **All matching sanctions entries** from official lists

These numbers balance relevance and noise reduction.

### D. Retrieval Timing

Retrieval occurs:

- at the start of reasoning
- after each major observation
- during conflict resolution
- before final synthesis

This ensures retrieval actively shapes reasoning.

## 5. Calibration Feedback Loop (Updating Thresholds Over Time)

A key refinement is enabling the agent to update its **calibration thresholds** such as false positive rates, false negative rates, and source reliability scores based on retrieval outcomes.

### How Calibration Data Is Stored

After each assessment, the agent stores:

- whether the final conclusion was later validated or contradicted

- which sources contributed to errors
- whether conflicts were resolved correctly
- whether retrieved facts were relevant or misleading

These are stored as semantic facts in the vector DB, tagged with:

- entity ID
- timestamp
- error type (FP, FN, borderline)
- contributing signals

**How Calibration Influences Future Reasoning**

During future assessments, the agent retrieves calibration facts and adjusts:

- source weighting
- conflict-resolution thresholds
- confidence scoring
- fallback strategies

Example:
If news-only sanctions claims produced false positives in the past, the agent reduces their weight in future conflict resolution.

This creates a **self-correcting retrieval-reasoning loop**.

**6. Retrieval-Related Failure Mode and Mitigation**

**Failure Mode: Semantic Drift / Irrelevant Retrieval**

A vector database may return semantically similar but irrelevant facts, such as confusing “Company X Holdings” with “Company X Manufacturing.”

**Mitigation Strategies**

1. **Entity Resolution Layer**
   Confirms identity match before using retrieved facts.
2. **Metadata Filtering**
   Filters by entity ID, jurisdiction, and date.

3. **LLM-Based Relevance Scoring**
   The LLM discards irrelevant retrievals.

**Calibration Feedback**
If a retrieval source repeatedly produces irrelevant matches, its weight is reduced.

**Conclusion**

Retrieval is essential for grounding, conflict resolution, historical continuity, and calibration. The agent integrates real-time retrieval, semantic vector memory, and a calibration feedback loop that improves accuracy over time. Careful design choices, data source selection, chunking, retrieval thresholds, and error-driven calibration ensure retrieval meaningfully enhances the agent's reasoning while mitigating risks such as semantic drift.

***Note**, I used Microsoft Copilot as a writing support tool to help structure and refine the language in this project description. All ideas, analysis, and final content decisions are my own.*